

May 26, 2025 10:00 PM GMT

Powering AI

The Inferences

The nexus of Asia's power demand and AI is getting stronger with datacenter investments set to double to US\$200bn and tighten power markets a lot faster. We turn more bullish on Asia's power requirements, and prefer power and natural gas infrastructure players as AI adoption spreads.

Key Takeaways

- #Inference 1: US\$200bn in potential Asia datacenter investment for 0.5 trillion more units of power by 2030, i.e., large as S. Korea's current power consumption.
- #Inference 2: Earnings rise 45% on avg. by 2027 for 27 key OWs for our Powering AI equities across India, Japan, China and Malaysia.



No AI without power: Power consumption continues to surprise our **bullish expectations** in Asia as AI adoption becomes more broad based, and tailwinds from on-shoring manufacturing and new age manufacturing sustain. We raise Asia's power consumption growth estimate by 30bps until 2030 (second increase in 2025) to reflect higher consumption in China, Japan, Malaysia, Thailand and now increasingly in India. Asia's power

consumption from DCs will rise to 100GW with a sixth of incremental power demand in Asia to be consumed by data centers. (Please contact us for our model).

How tight are power markets? Spot power prices have reflected lower coal/gas prices declining ~10% YTD, but market tightness is reflected in power price over fuel spreads that have widened in multiple geographies, with consensus margins for Asian power gencos rising 50bps YTD. A lot more gas is needed as multiple Asian countries are already looking at power shortages in the coming years and policy makers are racing to find solutions, with natural gas being top-of-mind. About a quarter of Asia (ex-China) DCs will be natural gas powered, while a large part of China's DCs will run on renewable power.

What's preferred? 1) Beneficiaries of "time to power" DCs are not yet fully appreciated by the market – especially grid operators. 2) Natural-gas-fired power plants, the most accessible means to dispatch power – can fill the power requirement void, especially as gas exports from the US rise; gas based pipeline suppliers also benefit as second derivative beneficiaries. And 3) infrastructure providers of clean power, grid equipment, cooling solutions and battery storage benefit. 2024 saw pricing power for power equipment companies, which we believe will now also be percolated to power generators, grid operators and midstream energy suppliers, such as gas pipeline

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companies: **Reliance Industries, Tenaga, CR Power, Gulf Development, Nari Technology, Mitsubishi Heavy – key picks.**

What surprised us in the past year? Policymakers have been calibrated in their approach to awarding more fossil-based power generation capacity and approving grid hardening investments, leading to only 7GW of new coal/gas/nuclear power generation capacity being announced in Asia ex China in the past year. Countries are re-starting existing power plants, either gas/nuclear, to keep power prices affordable and 10GW of restarts we believe could potentially be in the works.

Where have been the challenges? Slowdown in industrial demand for power as prices rise has been a key risk apart from availability of chips. In addition, multiple DCs in Asia are still being built for training, while inferencing demand for AI still needs to catch up.

Powering AI Equities: What's Priced in?

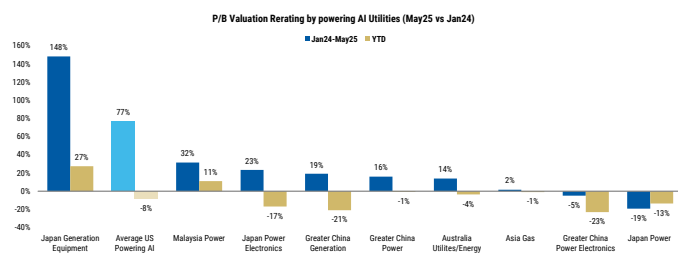
Data centres accounted for around 1.5% of the world's electricity consumption in 2024 and DC demand has been growing at 12% CAGR since 2017. US stocks powering AI equities have seen a -8% valuation multiple contraction YTD vs Asia peers between -23% to 27% despite robust AI demand and rising self-sufficiency. Dispersion has been wide within Asia with grid and hybrid power generators, specific equipment manufacturers and data center operators, all outperforming the local indices.

Natural gas is key to power AI: It is addressing the time to power as well as energy security. Gas share of generation will increase by 100bp to 10% in Asia and should drive 25mntpa of new gas consumption. Gas and renewable will work together to power the AI adoption curve.

What's priced in? Gas and power utilities with grid exposure in Asia outperformed benchmarks, but still de-rated YTD. Its interesting though that two year forward earnings expectations continue to rise as new DC capacity and overall power demand growth surprises in the region, and power margins have expanded. Stronger local currencies have also supported power utility earnings

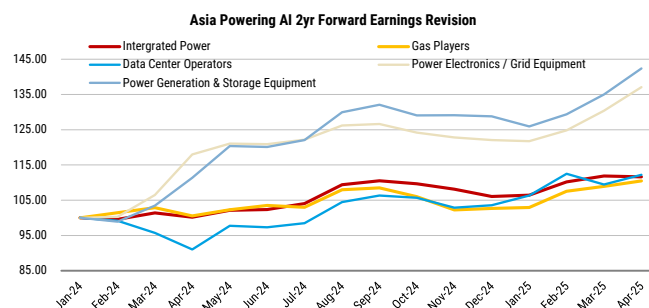
We see opportunities – especially in grid operators and power utilities – to play the theme as AI stocks come back into focus. Below we highlight our 37 'Powering AI' preferred picks in Asia for exposure to the theme across the value chain.

Exhibit 1: Asia Powering: AI beneficiaries have de-rated YTD...



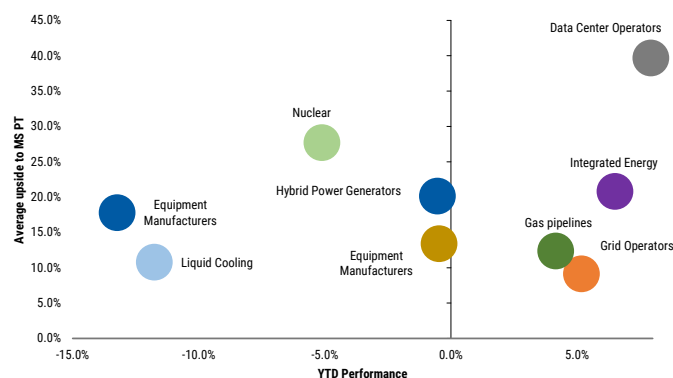
Source: Refinitiv, Morgan Stanley Research

Exhibit 2: ...despite an improved earnings outlook



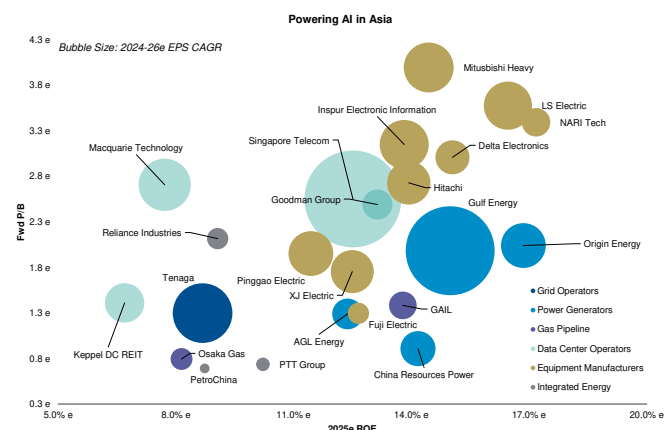
Source: Refinitiv, Morgan Stanley Research

Exhibit 3: Attractive themes in Asia Powering AI value chain – YTD performance vs average upside to our price targets



Source: Refinitiv, Morgan Stanley Research estimates, Priced at 26 May 2025

Exhibit 4: How Asia's Powering AI names stack up on valuations



Source: Refinitiv, Company data, Morgan Stanley Research estimates, Priced at 26 May 2025

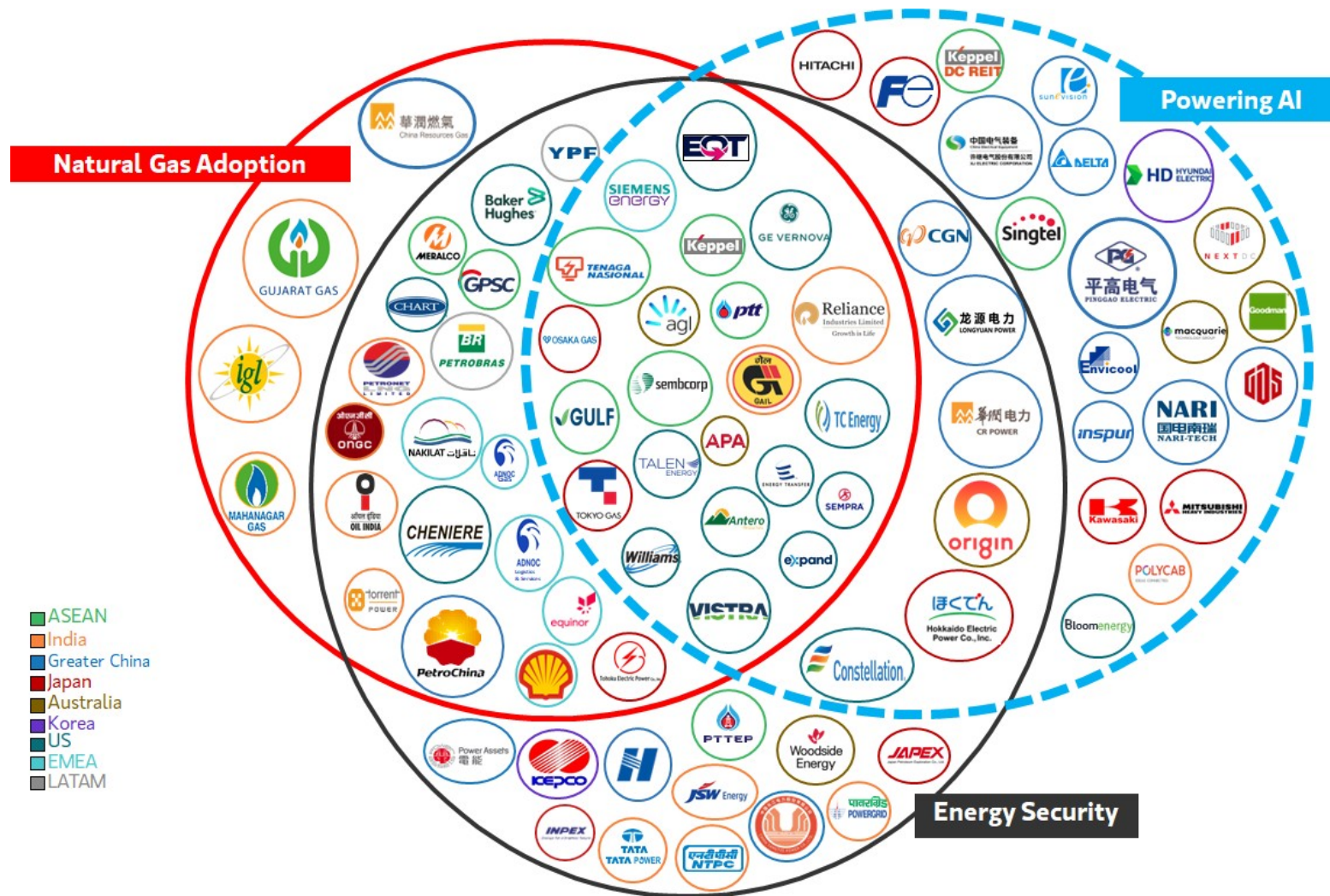
Exhibit 5:
Asia Powering AI-exposed stocks

Company Name	Ticker	Market cap, current, USD (MM)	3M ADTV, USD (MM)	Rating	Share price, last close	Price Target (Local CCY)	% Upside from last close	Country	Sub-Sector	Key Exposure Thesis	MS Analyst
Exposed to Powering AI Thematic											
Grid Operators											
Tenaga Nasional	TENA.KL	19,112	22	Overweight	14.00	16.30	16.4%	Malaysia	Grid Operator	Single grid operator benefits from doubling power demand	Mayank Maheshwari
APA Group	APA.AX	6,887	21	Equal-Weight	8.13	8.28	1.8%	Australia	Grid Operator	Bidding for new transmission line projects	Rob Koh
Power Generators											
Hokkaido Electric Power	9509.T	1,056	24	Overweight	706.20	960.00	35.9%	Japan	Integrated Power Utility	The company's service area is cooler hence making Hokkaido an advantageous location for AI datacenters	Reiji Ogino
Sembcorp Industries	SCIL.SI	9,073	21	Overweight	6.56	7.20	9.8%	Singapore	Hybrid Power	Tight electricity markets as Singapore expands DC capacity	Mayank Maheshwari
Koppel	KPLM.SI	9,587	23	Overweight	6.76	8.18	21.0%	Singapore	Hybrid Power	Tight electricity markets as Singapore expands DC capacity	Mayank Maheshwari
Gulf Development	GULF.BK	21,883		Overweight	48.00	69.00	43.8%	Thailand	Hybrid Power	Integrated DC + generation portfolio	Mayank Maheshwari
AGL Energy	AGL.AX	4,474	13	Equal-Weight	10.24	11.88	16.0%	Australia	Integrated utility	Leverage to power prices and renewable development	Rob Koh
CGN Power	1816.HK	3,635	22	Overweight	2.55 HKD	2.81 HKD	10.2%	China	Nuclear	Increased power demand from datacenters	Albert Li
China Resources Power	0836.HK	13,123	39	Overweight	19.84	23.70	19.5%	China	Hybrid Power	Increased Clean power demand from datacenters	Albert Li
Gas Pipelines											
GAIL	GAIL.NS	14,766	29	Overweight	191.28	248.00	29.7%	India	Gas Pipelines	Integrated Gas Player	Mayank Maheshwari
Tokyo Gas	9531.T	12,145	57	Equal-Weight	4,713.00	4,680.00	-0.7%	Japan	Gas Pipelines	Integrated Gas Player	Reiji Ogino
Osaka Gas	9532.T	9,985	28	Equal-Weight	3,614.00	3,910.00	8.2%	Japan	Gas Pipelines	Integrated Gas Player	Reiji Ogino
Data Center Operators											
NEXTDC	NXT.AX	5,452	21	Overweight	13.11	20.10	53.3%	Australia	Data Centers	Early DCs have >20% IRR, but higher land/development costs point to 10-12% ahead.	Andrew McLeod
Singapore Telecom	STEL.SI	49,916	93	Overweight	3.88	4.20	8.2%	Singapore	Data Centers	STEL's connectivity, power partnerships, and access to Nvidia's GPUs are key advantages to expand its AI DC footprint	Da Wei Lee
Goodman Group	GMG.AX	43,106	99	Overweight	32.68	37.50	14.7%	Australia	Data Centers	Global DC 3.9GW DC pipeline of secured power, plus ~1-2GW in procurement. Est AS40bn+ end product at 50%+ margins	Simon Chan
Macquarie Technology	MAQ.AX	1,023	4	Overweight	61.11	102.00	66.9%	Australia	Data Centers	High quality portfolio + an attractive pipeline of future DC re-investment opportunities	Andrew McLeod
GDS Holdings	GDS.O	5,540	86	Overweight	28.50 USD	42.00 USD	47.4%	China	Data Centers	China data center demand growth	Yang Liu
Koppel DC REIT	KEPE.SI	3,806	14	Equal-Weight	2.18	2.10	-3.7%	Singapore	Data Centers	Pure-play DC REIT	Derek Chang
SUNeVision	1686.HK	1,844	21	Overweight	6.12	10.00	63.4%	Hong Kong	Data Centers	Leading position in Hong Kong IDC market	Tom Tang
VNET Group	VNET.O	1,610	76	Overweight	5.99 USD	10.00 USD	67.1%	China	Data Centers	China data center demand growth	Tom Tang
Equipment Manufacturers											
Delta Electronics	2308.TW	32,291	103	Overweight	373.50	440.00	17.8%	Taiwan	Power Electronics	Key vendor of power electronics (~62% of total revenue) that improves electrical efficiency of data centers	Sharon Shih
Fuji Electric	6504.T	6,689	39	Equal-Weight	6,452.00	6,700.00	3.8%	Japan	Power Electronics	Power electronics demand growth	Masatoshi Terashi
Polycab India	POLC.NS	10,393	49	Overweight	5,883.50	7,352.00	25.0%	India	Wires and cables	Targets 10% export exposure by F26	Girish Achhipalia
Hitachi	6501.T	120,862	375	Equal-Weight	3,800.00	3,400.00	-10.5%	Japan	Power Grid Systems	Overseas sales of power grid systems	Masatoshi Terashi
NARI Tech	600406.SS	25,032	103	Overweight	22.45	26.51	18.1%	China	Power Grid Equipment	Power demand growth requires grid hardening capex	Eva Hou
XJ Electric	000400.SZ	3,047	61	Overweight	21.54	33.74	56.6%	China	Power Grid Equipment	Power demand growth requires grid hardening capex	Eva Hou
Pingao Electric	600312.SS	3,068	46	Overweight	16.29	20.85	28.0%	China	Power Grid Equipment	Power demand growth requires grid hardening capex	Eva Hou
HD Hyundai Electric	267260.KS	10,196	70	Overweight	386,000.00	460,000.00	19.2%	S. Korea	Power Grid Equipment	Power demand growth requires grid hardening capex	Ryan Kim
LS Electric	010120.KS	5,595	49	Equal-Weight	254,500.00	210,000.00	-17.5%	S. Korea	Power Grid Equipment	Power demand growth requires grid hardening capex	Ryan Kim
Mitsubishi Heavy	7011.T	75,152	1,325	Overweight	3,208.00	3,000.00	-6.5%	Japan	Gas Turbines	Higher demand for gas baseload drives turbine sales	Yoshinao Ibara
Integrated Energy											
PTT Group	PTT.BK	26,584	42	Overweight	30.50	34.80	14.1%	Thailand	Integrated Energy	Integrated Energy	Mayank Maheshwari
Reliance Industries	RELI.NS	224,599	215	Overweight	1,426.80	1,617.00	13.3%	India	Integrated Energy	Integrated Energy + expansion in new energy and AI DCs	Mayank Maheshwari
PetroChina	0857.HK	17,456	121	Overweight	6.48 HKD	8.75 HKD	35.0%	China	Integrated Energy	Integrated Gas Player	Jack Lu
Liquid Cooling											
AVC	3017.TW	7,781	127	Overweight	603.00	750.00	24.4%	Taiwan	Liquid Cooling	Liquid Cooling Exposure	Sharon Shih
Auras	3324.TWO	1,736	61	Equal-Weight	567.00	550.00	-3.0%	Taiwan	Liquid Cooling	Liquid Cooling Exposure	Sharon Shih
Envioco	002837.SZ	3,257	123	Overweight	31.50	37.50	19.0%	China	Liquid Cooling	Liquid Cooling Exposure	Chelsea Wang
Inspur Electronic Information	000977.SZ	9,946	354	Equal-Weight	48.67	50.00	2.7%	China	Liquid Cooling	Liquid Cooling Exposure	Howard Kao

Source: Refinitiv, Morgan Stanley Research, Priced at 26 May 2025

The Diffusion: Powering AI+ Natural Gas + Energy Security

Exhibit 6: Looking at our global Morgan Stanley Research coverage across multiple sectors, the gas power producers and electricity grid operators stand to benefit the most

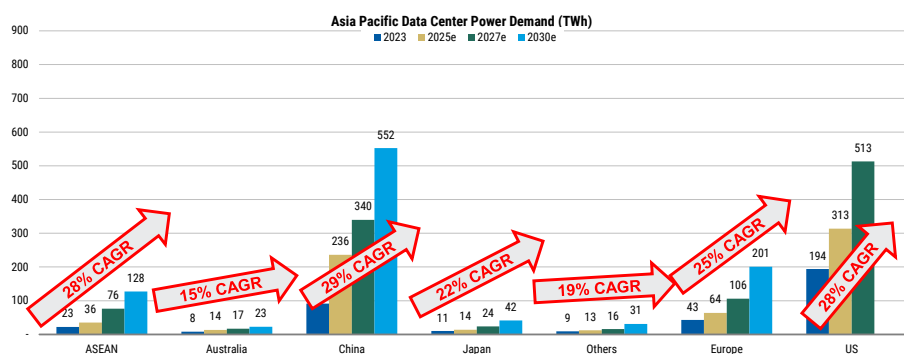


Source: Morgan Stanley Research

Data centers: A Quantum Expansion

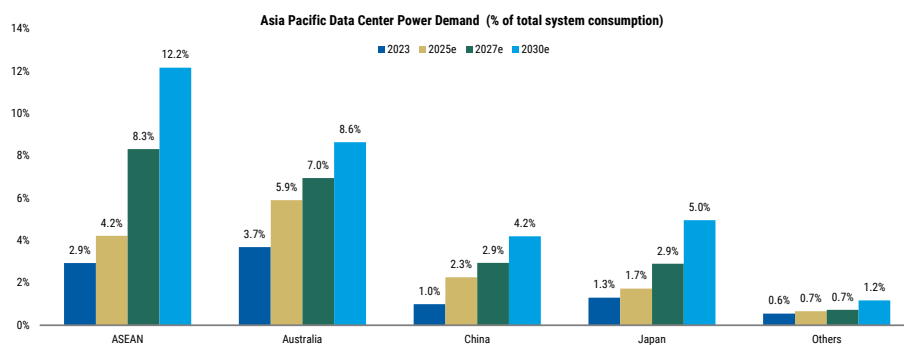
Power demand in Asia (including Japan) which includes data center capacity is set to nearly triple, to ~100GW by 2030, and form more than one-third of global data center capacity. Power demand growth is on track to double in Malaysia, Japan is reversing its multi-decade decline in power demand, and Australia, Taiwan and Singapore are seeing tighter power markets. We believe natural-gas-fired power plants – currently the most accessible means to dispatch power – are the most likely form of power generation that can fill the power requirement void apart from renewables (which also will need more natural gas or batteries to balance the grid). As the gas glut rises globally and prices fall to more competitive levels (Exhibit 18) compared to other alternatives, the upside across the gas value chain, including gas-fired power producers, should lead to upside risks in earnings and also a multiple re-rating. In China, while DCs will be 4% of total power consumption, a large part of it will be generated by renewables, similar to what we assume in Europe. However, DCs' renewable power adoption will raise gas-based generation requirements from other industries.

Exhibit 7: We estimate data center powering requirements nearly tripling into 2030 as the region sees ~70GW of DC capacity additions



Source: Morgan Stanley Research estimates

Exhibit 8: Data centers will be a significant portion of total power demand in southeast Asia and Australia, leading to even tighter power markets

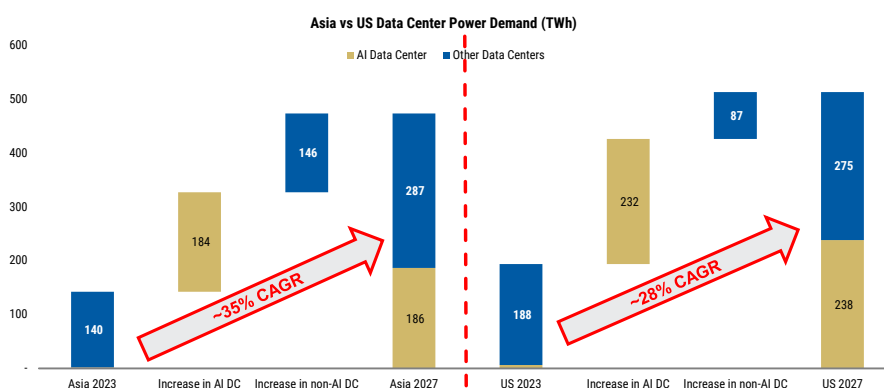


Source: Company Data, Morgan Stanley Research estimates

Asia's vs US datacenter power demand

Our bottom-up estimate for datacenter power demand in Asia is growing faster than the US driven by both AI and non-AI workloads. Renewed investor optimism on adoption also increased our AI inference demand forecasts in the region following the development of DeepSeek and the recent advancements of AI applications by Chinese companies.

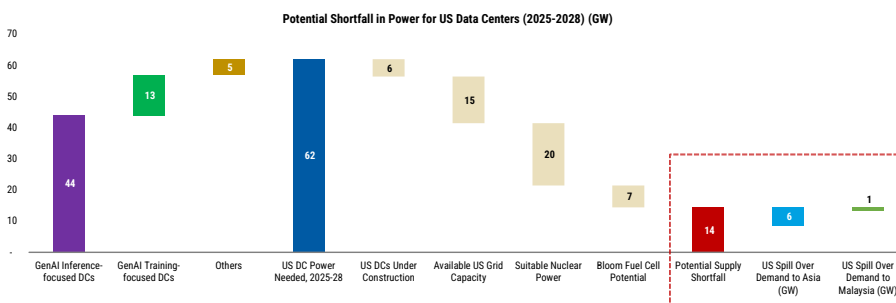
Exhibit 9: We estimate Asia's DC power demand CAGR of ~35% over 2023-27 exceeding that of the US, driven mainly by traditional DCs as well as AI DCs



Source: Company data, Morgan Stanley Research (e) estimates

US power supply chain constraints also help more AI capacity in Asia: We estimate a 14GW potential shortfall in power for US data centers by 2028 to have significant spillover effects for Malaysia and the rest of Asia ([Exhibit 15](#)). Our US power team estimates 57GW of incremental US GenAI-focused DC demand with ~44GW of inference-focused DCs implying 13GW of training-focused DCs. With US power grids near capacity, we believe at least 6GW of training workload could be off-shored to Asia by end of the decade to unlock the "Time to Power" value from the [US\\$1 trillion GenAI economy](#). We estimate incremental power demand would be driven by gas as the region remains committed to transition away from coal-based generation and the ramp up of nuclear capacity proceeds gradually.

Exhibit 10: We estimate ~14GW of potential shortfall in power for US DCs factoring suitable nuclear and fuel cell powering potential

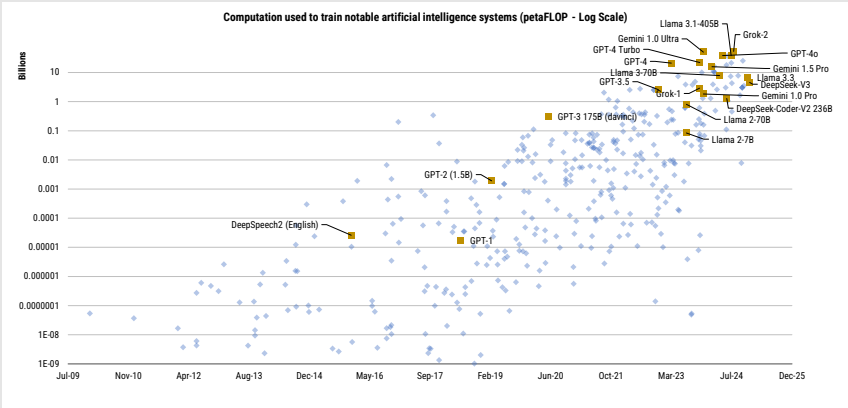


Source: Company Data, Morgan Stanley Research estimates

More efficient models to drive increased AI adoption across more use cases – Recent advances in LLM by DeepSeek R1 have demonstrated that cutting-edge AI

capabilities can be achieved with significantly less hardware than other iterations and have defied conventional expectations of computing power requirements. We believe if the adoption curve inflects (as tech companies accelerate investments), especially as MS Tech teams sees the commoditization and commercialization of foundation models ([see here](#)), we think the long-term growth estimate of 4%+ power demand could see upside risks.

Exhibit 11: Evolution of AI training compute requirements shows a ~10x increase in compute for every generation of LLM



Source: Epoch, OurWorldInData, Morgan Stanley Research

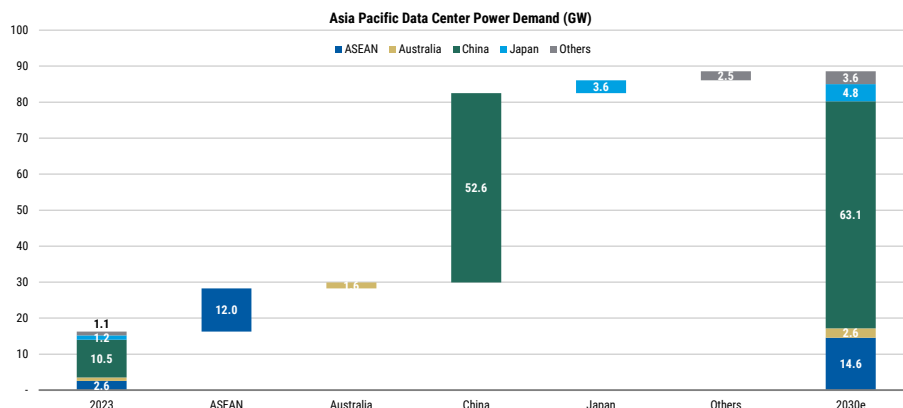
Exhibit 12: Global Data Center Clusters



Source: IEA

Who is adding DC capacity in Asia?

Exhibit 13: China will drive a significant portion of data center growth in Asia



Source: Company Data, Morgan Stanley Research estimates

China: We estimate that the top six internet and cloud service providers (CSPs) in China will increase their annual capex by 59%, to Rmb373bn (US\$50bn+) in the next three years. We project this to translate into average new data center booking demand of 3-4GW p.a. in the next three years – a 20% CAGR. We estimate China's AI GPU self-sufficiency ratio will rise from 34% in 2024 and to 82% in 2027, which will aid adoption and pace of rollout. We estimate DC power supply to rise from 2% of total demand to 4% by 2030. China aims to supply 100% of DC power demand from green energy by the early 2030s vs. ~20% in 2023, and it requires green power to be >80% of the mix for newly built national hub data centers by the end of 2025. [See our views in Aligning AI with Green Energy and Smarter Grids](#) and [China's Answer to Powering AI](#)

Southeast Asia: Malaysia remains a primary beneficiary of the buildout of data centers in Southeast Asia with Tenaga's pipeline growing to ~6GW (two thirds of Southeast Asia's DC pipeline). We also see Thailand beginning to add capacity in a significant way with over US\$10bn of commitment by hyperscalers to develop AI and cloud infrastructure. In Southeast Asia, we prefer Tenaga to play the theme. Tenaga's DC pipeline remains strong. Load utilization in Feb-25 was 700MW (~37% on 1.9GW completed DC capacity) vs 400MW in Dec-24. Tenaga said it has not seen any slowdown in the 6GW (38 projects) DC pipeline and has ~39 additional projects with submitted applications, and ~60 at pre-consultation. Only two of the 39 DCs at the application stage are AI-focused with no signs of withdrawal of existing projects, per Tenaga.

Japan: According to the Japanese government's 7th Strategic Energy Plan, Japan's electricity demand is expected to increase to 11.1-1.2trn kWh in FY2040 (F3/41), from 987.7bn kWh in F3/24). The main factor driving the increase in electricity demand is the construction and expansion of AI data centers. AI data center growth should spur related electric power business; industries likely to benefit are electric utilities, wire & cable, machinery and capital goods, construction, and telecommunications. [See our views in Powering AI in Japan \(2025\)](#)

India: India has 2GW of datacenter capacity with plans to grow that to 5GW by 2030. Reliance is now building Gen AI infrastructure to provide to hyperscalers along with tie up

with NVIDIA for chips for “Large Scale Internal Adoption of AI” and embedding AI in all of the group’s processes. We estimate one GW datacenter facility would need ~ 678k B100 chips and if RIL were to use ~200MW for its own purposes it would need ~135k B100 chips (details are limited). Also 1GW of datacenter capacity when scaled up, which normally needs 4-5 years from startup, would need ~1.3GW of round-the-clock power and we think RIL’s new energy ecosystem should be able to contribute to this. RIL also has a 33.3% stake in a partnership with Brookfield to build datacenter capacity with current .

Australia: Electricity load growth from data centers, among other things, adds pool price support and customer opportunity (e.g., Energy Hub) – something we haven’t seen in the National Electricity Market for a decade. Electricity markets remain tight with the percentage of expected unserved energy above the reliability standard in the key states of New South Wales and Victoria, and we only expected to rise despite the increase in capacity and grid investments committed by the regulators.

Korea: South Korea’s finalized 11th Basic Plan for Long-Term Electricity Supply and Demand (BPLE), released in February 2025, underscores the nation’s strategic response to the escalating electricity demand from data centers, AI, and semiconductor sectors. The plan projects a significant increase in data center power demand, revising the estimate from 1.4GW in the 10th plan to 4.4GW by 2038. As of 2024, South Korea’s data center power capacity stands at around 765MW. Projections indicate this will increase to approximately 2GW by 2029, reflecting the growing demand for digital infrastructure.

~US\$200bn of Committed Investments

Exhibit 14: ~US\$200bn in notable investments in Asia Cloud and AI Infrastructure, with upside, in our view

Company	Value (US\$ bn)	Investment Scope
Malaysia		
Amazon Web Services (AWS)	6.0	Expand AI & cloud infrastructure by 2037
AirTrunk	NA	JHB1 DC, 50MW initially by 2024
Microsoft	2.2	Expand AI & cloud infrastructure 2024-28
Google	2.4	Expand AI & cloud infrastructure by 2035
Nvidia - YTL Power	4.3	AI Cloud and Supercomputer Infrastructure
GDS Holdings	2.0	Johor Data Center Expansion
ST Telemedia Global DC	1.0	Johor Data Center Expansion
NTT Communications	NA	290MW campus (68.5acres), 48MW by Apr-27
Yondr Group	NA	200MW campus (72.8acres at Johor STP)
Bridge DC	NA	110MW campus at Johor STP
PDG	NA	150MW campus (31acres at Johor STP), 60MW by 2Q24
Thailand		
Microsoft, Google	3.5	Expand AI & cloud infrastructure
Amazon Web Services (AWS)	5.0	Expand AI & cloud infrastructure by 2037
Gulf Energy	0.1	Data Center Expansion with AIS and Singtel
GDS Holdings	0.8	Hyperscale DC Project operational by 2027
Darmac	1.0	3-4 each 20MW DC Projects
TikTok	8.8	Datacenter and Cloud infrastructure
Indonesia		
Microsoft	1.7	Expand AI & cloud infrastructure 2024-28
GDS Holdings	NA	Batam Data Center Expansion
Singapore		
Amazon Web Services (AWS)	8.6	Expand AI & cloud infrastructure by 2028
Singtel	1.0	~90MW Advanced Data Center pipeline in SG,TH&MY
Google	4.2	Increase Investments to US\$5bn with 4th DC
Singapore Government	0.7	Govt. comits ~US\$700 in AI Development
ASEAN Total	53.2	
Japan		
Amazon Web Services (AWS)	15.0	Expand cloud infrastructure from 2023-27
Microsoft	2.9	AI & cloud infrastructure 2024-26
Oracle	8.0	Expand cloud infrastructure by 2034
Google	1.0	Boost connectivity to Japan via two subsea cables
Australia		
Amazon Web Services (AWS)	5.0	Expand cloud infrastructure in Sydney through 2027
Microsoft	1.0	96MW Sydney Data Center
Microsoft	2.2	Expand AI & cloud infrastructure by 2026
NEXTDC	NA	550MW Sydney DC
Greater China		
GDS	0.4	Data Center Expansion in Hong Kong
Alibaba	52.0	AI infrastructure
Tencent	13.0	Increased annual cloud capex
ByteDance	22.0	Increased annual cloud capex
Baidu	1.4	Increased annual cloud capex
Kuaishou	2.0	Increased annual cloud capex
Meituan	1.5	Increased annual cloud capex
Greater China	92.3	
India		
Reliance	NA	1GW AI ready DC partnering with Nvidia
TATA Communications	NA	AI ready DC partnering with Nvidia
Amazon Web Services (AWS)	4.0	Expand cloud infrastructure
Princeton Digital Group	1.0	AI ready DC with total capacity of 230 MW
ST Telemedia Global DC	1.0	Data Center Expansion in India
Google	NA	Sovereign AI cloud and Gemini 1.5 Flash AI model DC
Microsoft	3.5	Cloud and Gen AI
India Total	10.0	
Asia Others		
Keppel DC REIT	2.0	Data Center Expansion in China and Keppel Pipeline
Asia Total	192.6	

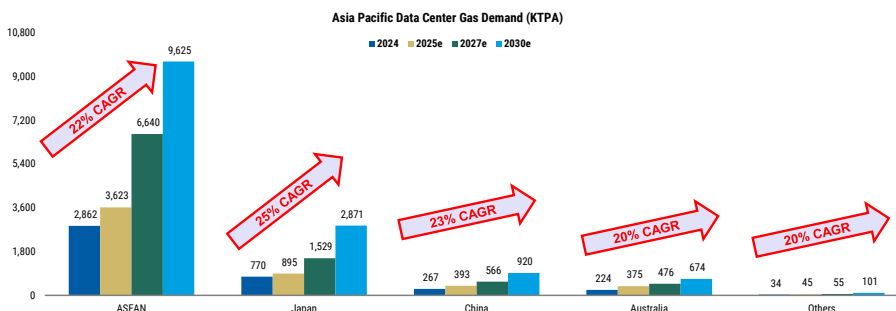
Source: Company Data, Newsflow, Morgan Stanley Research

AI: Powered by Gas

"Data center and artificial intelligence booms, along with additional coal retirements, expected to drive a further ~10 Bcf/d of incremental natural gas demand by 2030", EQT Corporation, 24 March, 2025

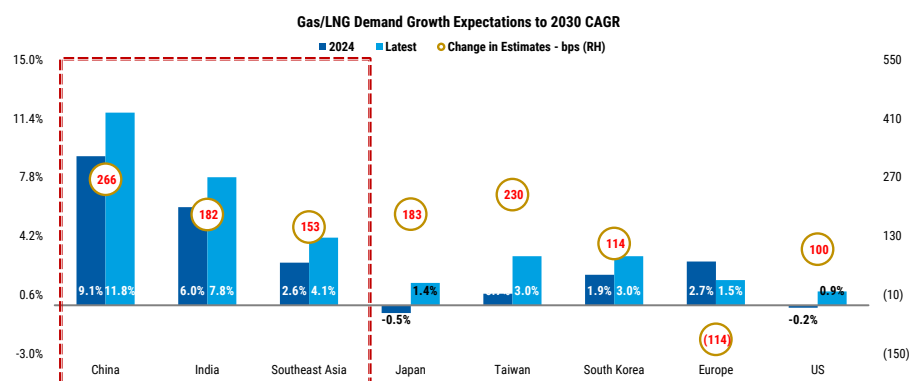
We estimate 14mntpa of natural gas requirements to power one-tenth of Asia Pacific's new data center capacity (one-fifth ex-China) and 6% of incremental Asia's gas needs would be driven by data center needs for power in India, Japan and South East Asia, though a large part of China's demand for data centers will be fulfilled by renewables.

Exhibit 15: We think natural gas plays a key role in meeting APAC's growing appetite for power, with data centers potentially adding ~14mntpa of incremental gas demand by 2030



Source: Morgan Stanley Research estimates

Exhibit 16: Our global gas demand estimates have been revised upwards by more than 100bp in most regions



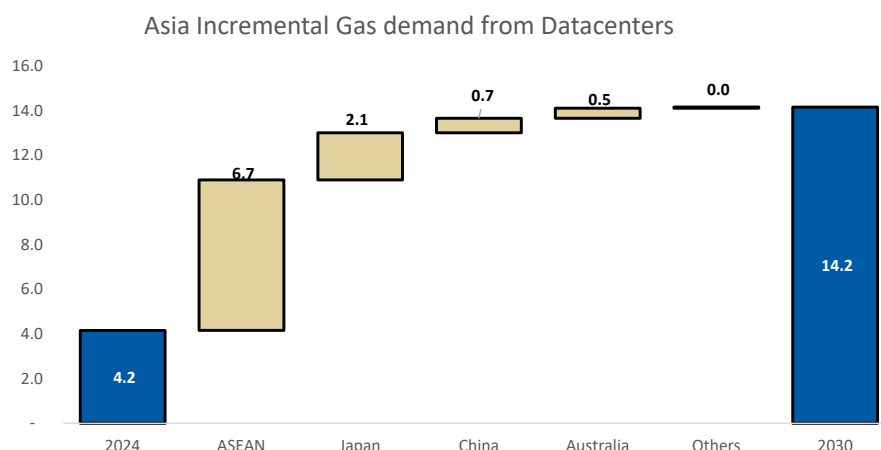
Source: Morgan Stanley Research Estimates

We estimate gas demand from data centers will add ~14mntpa by 2030 to Asia's natural gas consumption (Exhibit 17), with India, Malaysia, Singapore, and Taiwan followed by Japan taking the lead in meeting their data center and semiconductor industry electricity requirements via gas-based power. Nuclear, in our opinion, along with renewables (Exhibit 74), will have a role to play in the growing AI-driven power consumption, especially in China and Japan. That said, investors are under-appreciating the time to power for nuclear fuel, in our view, and hence gas-based power offers a good

alternative from an energy security perspective.

According to the Japanese government's Seventh Basic Energy Plan, Japan's electricity demand is expected to increase to 1.1 to 1.2 trillion kWh (987.7 billion kWh in F24) in 2040, largely due to AI data centers. According to the plan, nuclear power generation and renewable energy generation (solar and wind) will increase in 2040. However, we believe that there is little possibility of a significant increase in solar and wind power generation because its cost is high. In the long term, we believe that gas-fired power generation will increase instead of renewable energy generation. Increasing gas-fired power generation means increasing demand for natural gas in the country.

Exhibit 17: South East Asia, followed by Japan, will drive the first leg of gas power demand growth, making up over half of the incremental 14mntpa of data center-led gas power demand



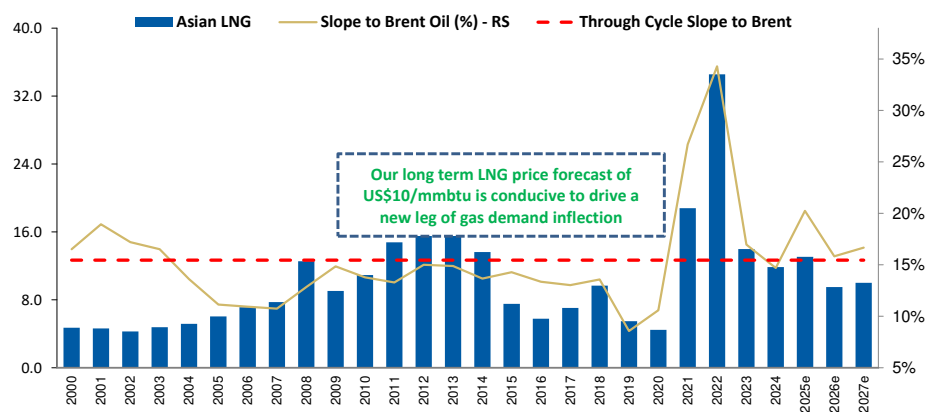
Source: Morgan Stanley Research estimates

Gas – A reliable lower carbon base load for data center's 24/7 power requirements:

Renewables alone can't service DC power demand due to intermittency. Gas-based power capacity has scope to grow its utilization rate and support the 24/7 power demand of data centers. Gas especially can be combined with renewables (solar/wind generation in Asia) to reduce carbon intensity (while also providing quality power stability) at competitive prices. While the increased need for power should benefit power producers like Sembcorp Industries in Singapore, it also provides medium-term upside to gas pipeline infrastructure companies amid the backdrop of a global gas glut.

Gas Solves the Powering AI Trinity: Time to Power, Safety and Reliability: While nuclear power remains the ultimate source of clean and reliable base load, the time required to setup a new nuclear power plant remains 8-10 years with significant regulatory hurdles. This is while demand for data centers is ever present today and hyperscalers are willing to pay between 20-30% premium for immediate power supply connection. Gas power plants require less permitting and it is a proven, well-accepted power generation technology which solves speed to market, safety, and reliability concerns, in our view.

Exhibit 18: Gas Competitiveness: Asia LNG prices at US\$10/mmbtu appears attractive to catalyze demand in price sensitive regions of India, South East Asia



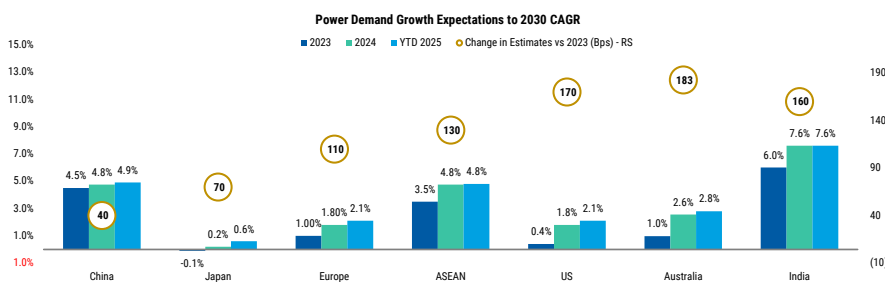
Source: BP stats, Refinitiv, Morgan Stanley Research (e) estimates

Power Demand is Inflecting Globally

Power consumption is at another tipping point after tripling since 1980. Over 25 trillion units (kwh) of power was consumed worldwide last year, and we see this growing rapidly in coming years. Of this, APAC consumed over 13 trillion units (kwh). We forecast APAC power consumption (Exhibit 21) will grow ~5% CAGR through 2030, 25% faster than the previous decade, with gas meeting 10% of incremental power demand supporting renewables and nuclear (Exhibit 23).

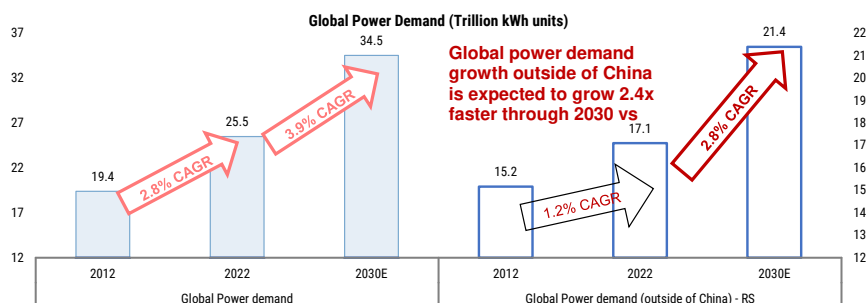
The boom in power demand is driven by multiple levers, in our view, such as electrification, supply chain localization, and data center power demand. Excluding weather effects on power demand, we have seen industrial and commercial sector demand for power rise faster than ever since 2012 in India, the US and Malaysia, and markets such as Indonesia, Australia, Japan and Taiwan are catching up. The tightness in supply has been acknowledged by policy makers with multiple countries extending the life of base load power plants and some restarting coal-fired generation.

Exhibit 19: Our power demand expectations have been revised up globally



Source: Morgan Stanley Research estimates

Exhibit 20: We expect power demand to inflect globally driven by data centers, new supply chains and electrification of industry



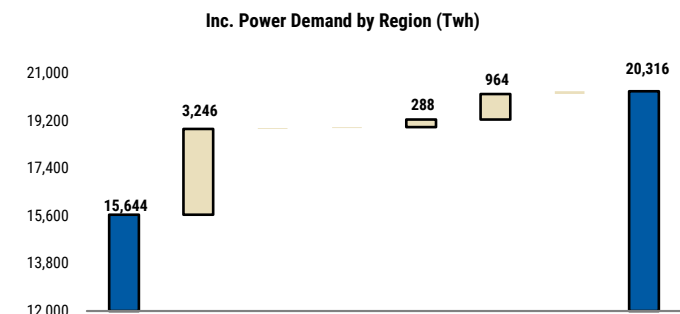
Source: Morgan Stanley Research estimates

Demand shifts higher: We forecast global demand at 3,400TWh by 2030 but estimate generation from renewables (including hydro) will only provide half of this demand, leaving a lot to be produced using fossil fuel or nuclear sources. Increasing power demand from data centers, a pick-up in industrial activity in large markets due to near-shoring, and weather are all playing a part in rising power requirements. We have seen an average 80bps increase in forecast consensus growth rates across key power demand geographies through 2030.

What is driving the demand in Asia? We estimate APAC's power demand will grow at

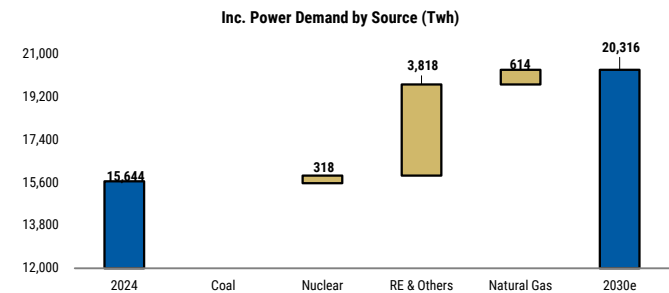
~4.7% CAGR into 2030. China and India will see the fastest growing regions followed by South East Asia. Ex-China, we estimate power demand to grow at ~4.3% CAGR. Renewables and natural gas will underpin power consumption growth and steadily rise in the region's electrification mix, at the expense of coal-based power, which we think will be flattish into 2030. Renewables-based power will likely grow the fastest, led by capacity growth and modest efficiency improvements. Nuclear power (led largely by China and Japan) and natural gas will also see meaningful growth with the rollout of new capacities, favorable economics and higher operating rates for gas.

Exhibit 21: We estimate APAC's power requirements will rise ~30% by 2030 led by China and India



Source: Morgan Stanley Research estimates (e) based on data from EPPO Thailand, ESDM Indonesia, MEIH Malaysia, EMA Singapore, PetroVietnam, EIA, PNGRB, Morgan Stanley Research

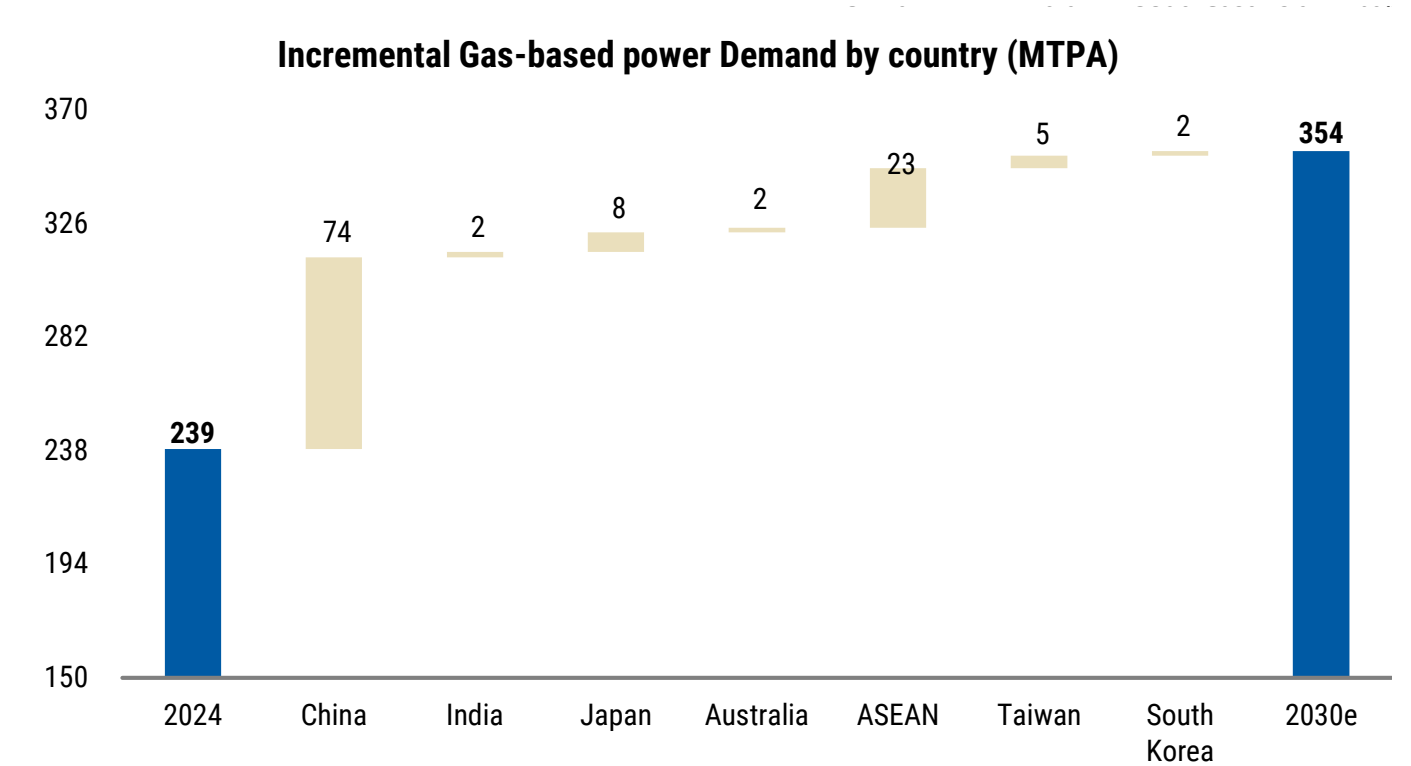
Exhibit 22: While renewables (largely in China) will make up two-thirds, natural gas will play a vital role in meeting this demand



Source: Morgan Stanley Research estimates (e) based on data from EPPO Thailand, ESDM Indonesia, MEIH Malaysia, EMA Singapore, PetroVietnam, EIA, PNGRB, Morgan Stanley Research

Exhibit 23:

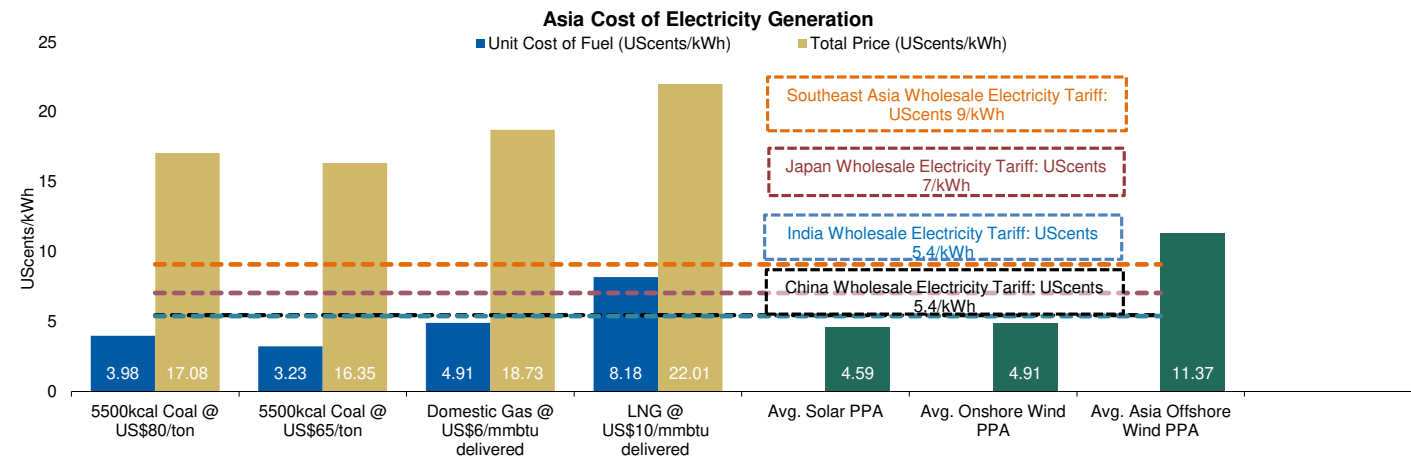
Need for cleaner power sources in China, ~10GW of new gas-based capacities in South East Asia and higher run rates of stranded gas-based power in India will underpin gas demand



Source: Company Data, Morgan Stanley Research estimates

LNG's competitiveness vs. alternative fuels like coal, domestic gas, propane and fuel oil has been a fundamental driver for demand ([Exhibit 24](#)). Overall at US\$9-10/mmbtu of delivered price, i.e. near the total cost of US Henry Hub gas delivered into Asia (including liquefaction), while less attractive vs. US\$80/ton of Indonesian coal, it should be able to complement renewables and be competitive vs. alternatives like propane.

Exhibit 24: Cost of domestic gas-based electricity generation in Asia is almost near coal parity, while imported LNG-based electricity can be used for peak loads and in conjunction with renewables



Source: Company Data, Morgan Stanley Research estimates

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(as of April 30, 2025)

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Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	603	16%	76	9%	13%	229	13%
Total	3,774		828			1717	

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Industry : S. Korea Technology



Stock Rating History: 5/1/20 : NA/A; 7/19/21 : NA/I; 8/12/21 : NA/C; 10/4/22 : NA/A; 7/21/24 : NA/I; 9/15/24 : NA/C; 3/25/25 : O/C

Price Target History: 3/25/25 : 460000

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Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

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Industry : S. Korea Technology



Stock Rating History: 5/1/20 : NA/A; 7/19/21 : NA/I; 8/12/21 : NA/C; 10/4/22 : NA/A; 7/21/24 : NA/I; 9/15/24 : NA/C; 3/25/25 : E/C

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INDUSTRY COVERAGE: ASEAN Utilities and Infrastructure

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/26/2025)
Mayank Maheshwari		
Aboitiz Power Corporation (APPS)	U (06/30/2021)	PP35.50
Airports of Thailand (AOT.BK)	O (08/25/2021)	Bt34.00
Global Power Synergy PCL (GPSC.BK)	O (02/08/2024)	Bt33.50
Gulf Development PCL (GULF.BK)	O (03/26/2025)	Bt47.00
Manila Electric Company (MER.PS)	O (06/20/2022)	PP536.00

Perusahaan Gas Negara (PGAS.JK)	U (05/17/2023)	Rp1,815
SembCorp Industries Ltd (SCIL.SI)	O (06/09/2020)	S\$6.65
Tenaga Nasional (TENA.KL)	O (09/12/2023)	RM14.00
Vivek Rajamani		
International Container Terminal Service (ICT.PS)	O (03/04/2024)	PP405.60

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* Historical prices are not split adjusted.